

Gear Ratio & Gear Train Simulation Lab

Name: _____

Period: _____

Date: _____ Gear pair investigation

PART A

Predict Before You Simulate

Use the simulation to test your predictions. Do not change an answer just because your first prediction was wrong - record what you learned.

Gear ratio: $GR = \text{driven teeth} / \text{driver teeth}$

Output speed: $\text{rpm}_{\text{out}} = \text{rpm}_{\text{in}} \times (\text{driver} / \text{driven})$

Ideal torque: $T_{\text{out}} = T_{\text{in}} \times (\text{driven} / \text{driver})$

External mesh: direction reverses at each mesh

For every two-gear external mesh, predict the driven gear behavior before opening the simulation.

System	Predict speed	Predict torque	Predict direction	Simulation check
12T driver -> 36T driven				
40T driver -> 20T driven				
24T driver -> 24T driven				

What pattern do you notice?

Vocabulary in your own words

Driver gear: _____

Driven gear: _____

Gear ratio: _____

Torque: _____

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PART B

Explore Gear Pairs

Run at least four trials. Include one reduction, one speed-up, one 1:1 pair, and one system of your choice. Record the live values shown by the simulation.

Suggested starting values: 600 rpm input speed and 0.50 N*m input torque.

Trial	Driver	Driven	GR	Input rpm	Output rpm	Input T	Ideal Output T	Direction
1								
2								
3								
4								

1. Which trial produced the greatest ideal output torque? Explain using the gear ratio.

2. Which trial produced the greatest output speed? What happened to ideal torque?

3. A rover wheel needs slow, strong rotation. Would you choose a gear ratio greater than 1, less than 1, or equal to 1? Why?

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PART C

Build a Gear Train

Load the 20T -> 24T -> 40T example. Then change ONLY the idler gear tooth count. Keep the first driver and final output gear unchanged.

Key question: Does changing the idler tooth count change the overall ratio?

Track ratio and direction separately. Count external meshes for direction.

Idler	Overall GR	Input rpm	Output rpm	Input direction	Output direction
12T		500		CW	
24T		500		CW	
36T		500		CW	
60T		500		CW	

4. What stayed the same as the idler tooth count changed?

5. What changed visually when the idler size changed?

6. Explain why the idler does not change the overall ratio in this simple gear train.

Direction Challenge

Build a three-gear train whose final output rotates in the SAME direction as the driver. Record one working tooth-count combination.

Driver: _____T Idler: _____T Output: _____T Input direction: _____ Output direction: _____

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PART D

Lunar Sample-Storage Door

The motor runs efficiently near 1200 rpm. The door should rotate slowly and gain ideal torque so it is less likely to stall in lunar dust. Compare the three options in the simulation.

Assume input torque = 0.50 N*m for comparison. All options use one external mesh.

Option	Gear pair	GR	Output speed	Ideal output torque	Direction
A	12T -> 24T				
B	12T -> 48T				
C	24T -> 36T				

7. Which option do you recommend for the storage door?

8. Use at least TWO numerical pieces of evidence to justify your recommendation.

9. Identify one real-world factor that could make the actual output different from the ideal calculation.

EXIT CHECK

10. What happens to output speed and ideal torque when a small driver turns a much larger driven gear?

11. What is the purpose of an idler gear in a simple external gear train?

12. How does each external gear mesh affect rotation direction?
